

BFS Traversal

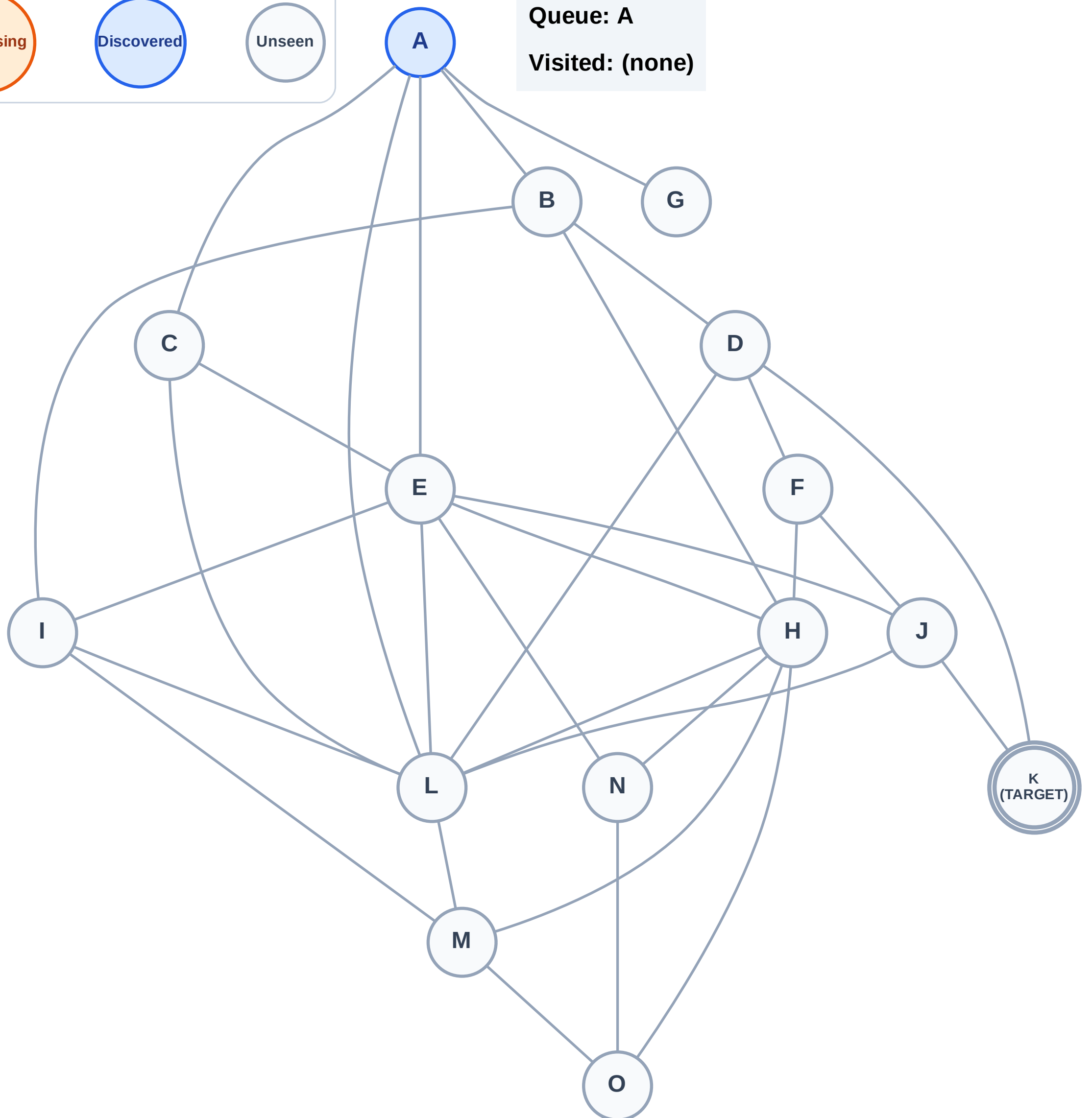
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

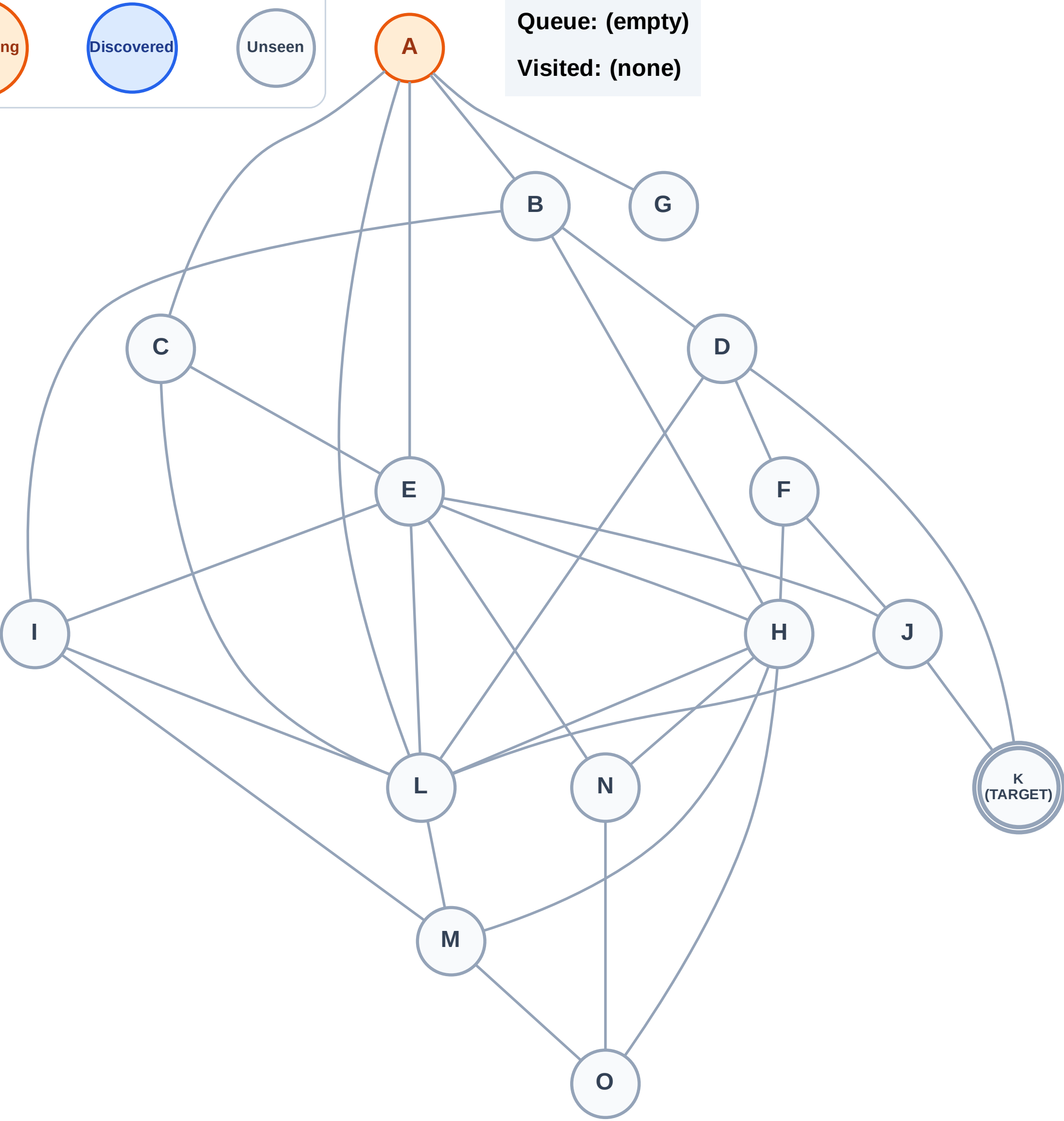
Complete

Processing

Discovered

Unseen

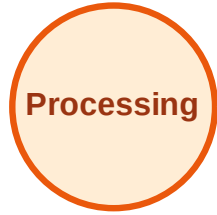
Queue: (empty)
Visited: (none)



BFS Traversal

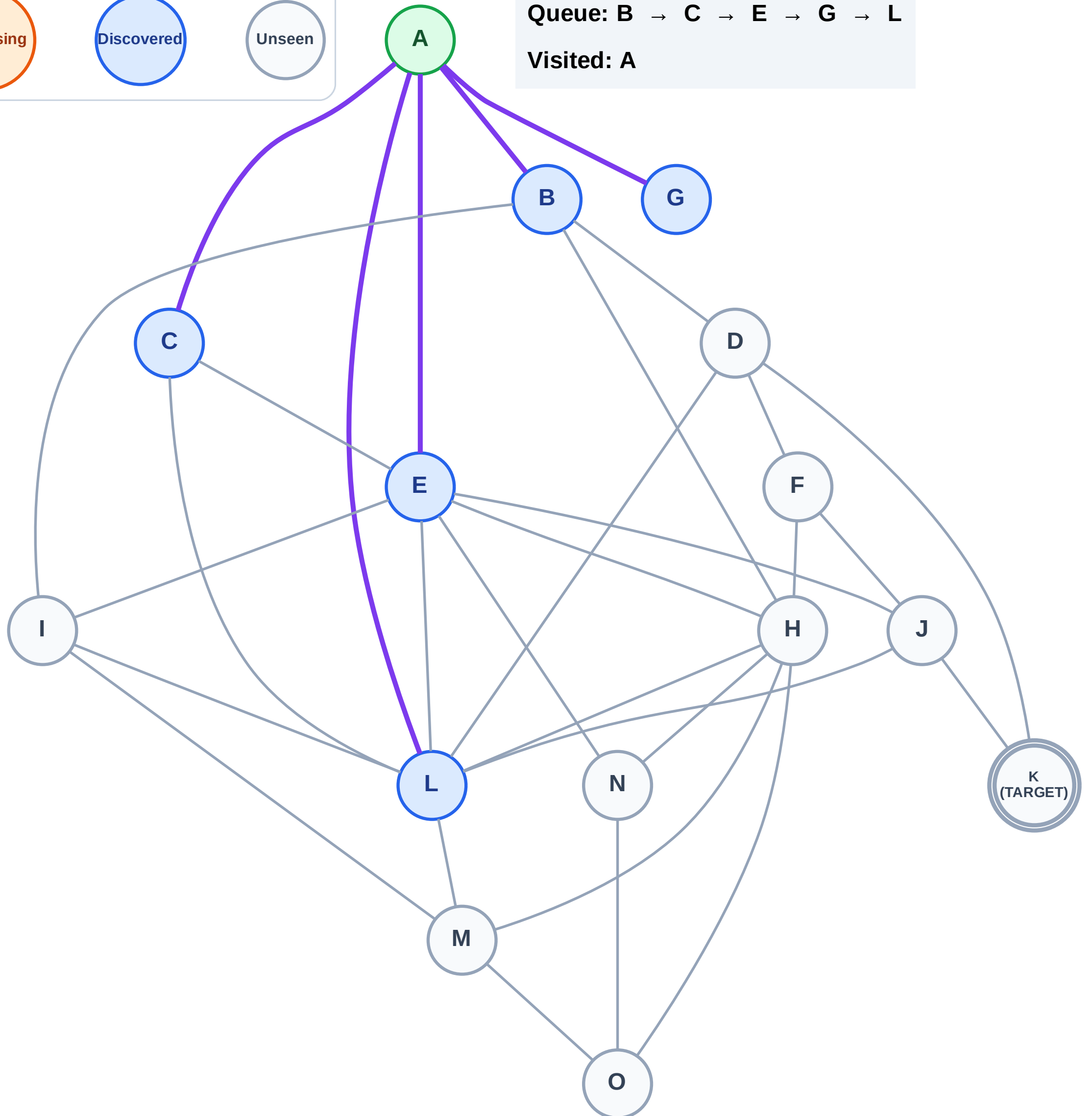
Step 3: Discover B, C, E, G, L from A

Legend



Queue: B → C → E → G → L

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

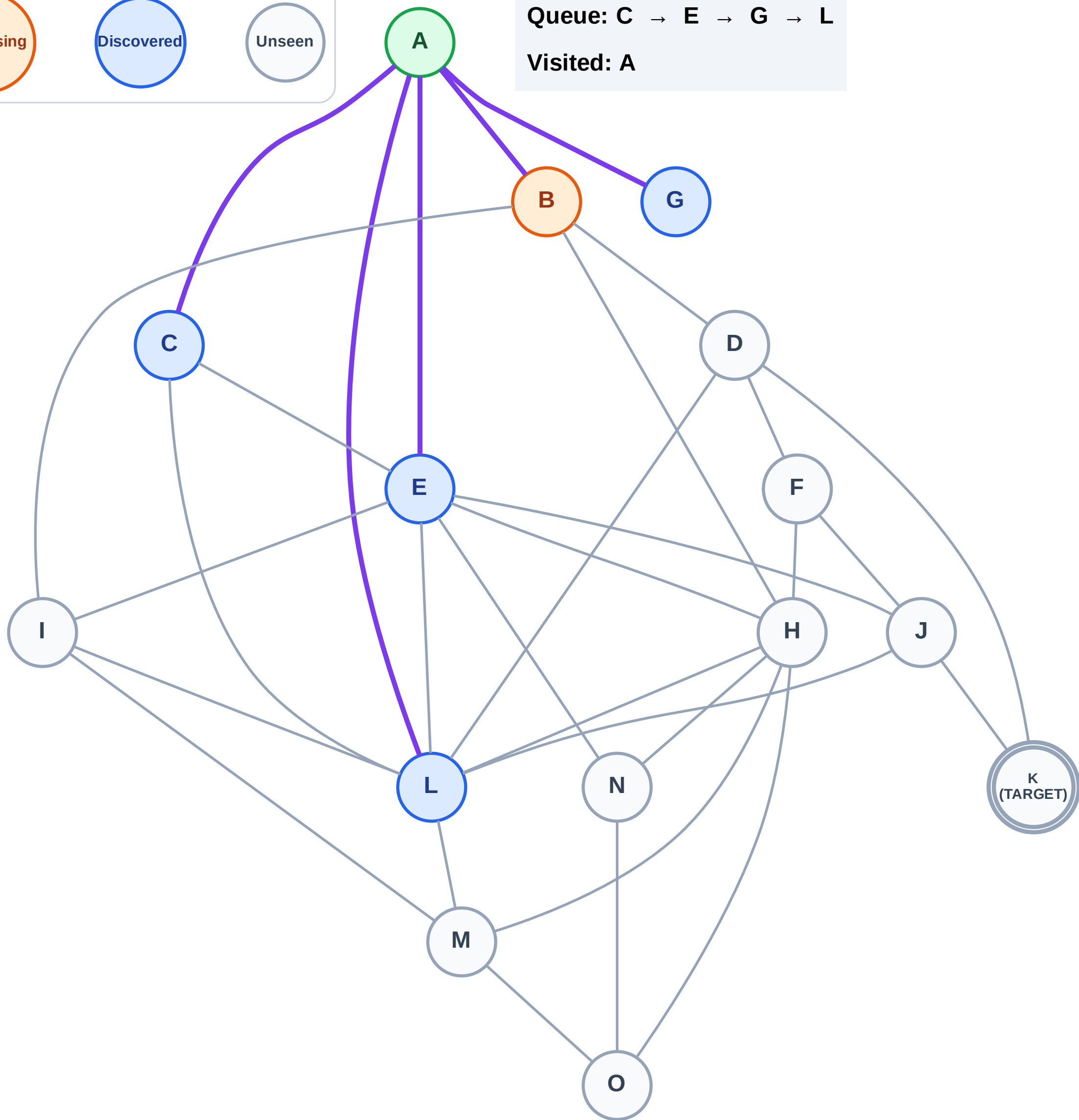
Processing

Discovered

Unseen

Queue: C → E → G → L

Visited: A



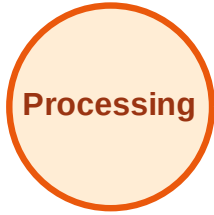
BFS Traversal

Step 5: Discover D, H, I from B

Legend



Complete



Processing



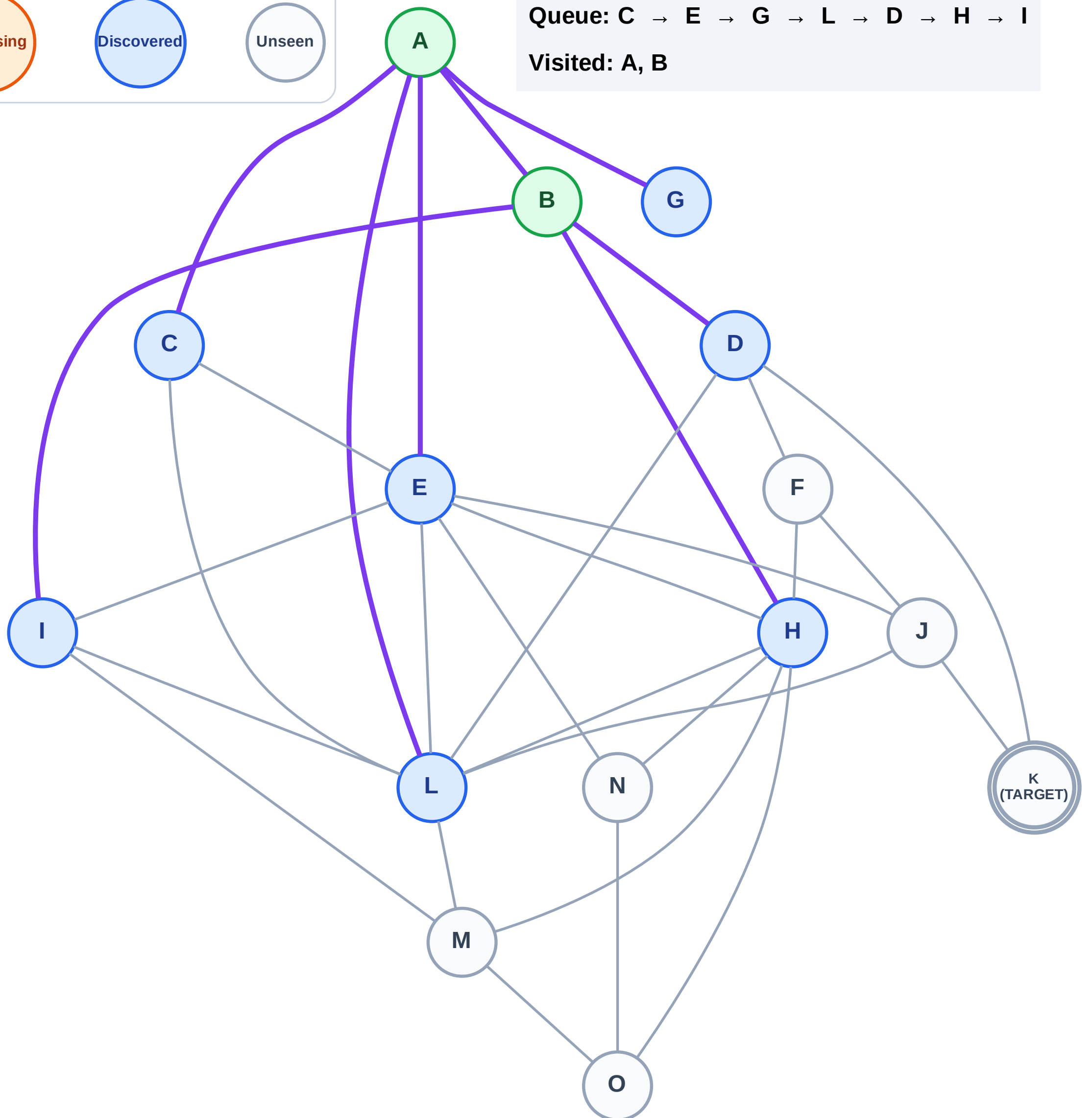
Discovered



Unseen

Queue: C → E → G → L → D → H → I

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

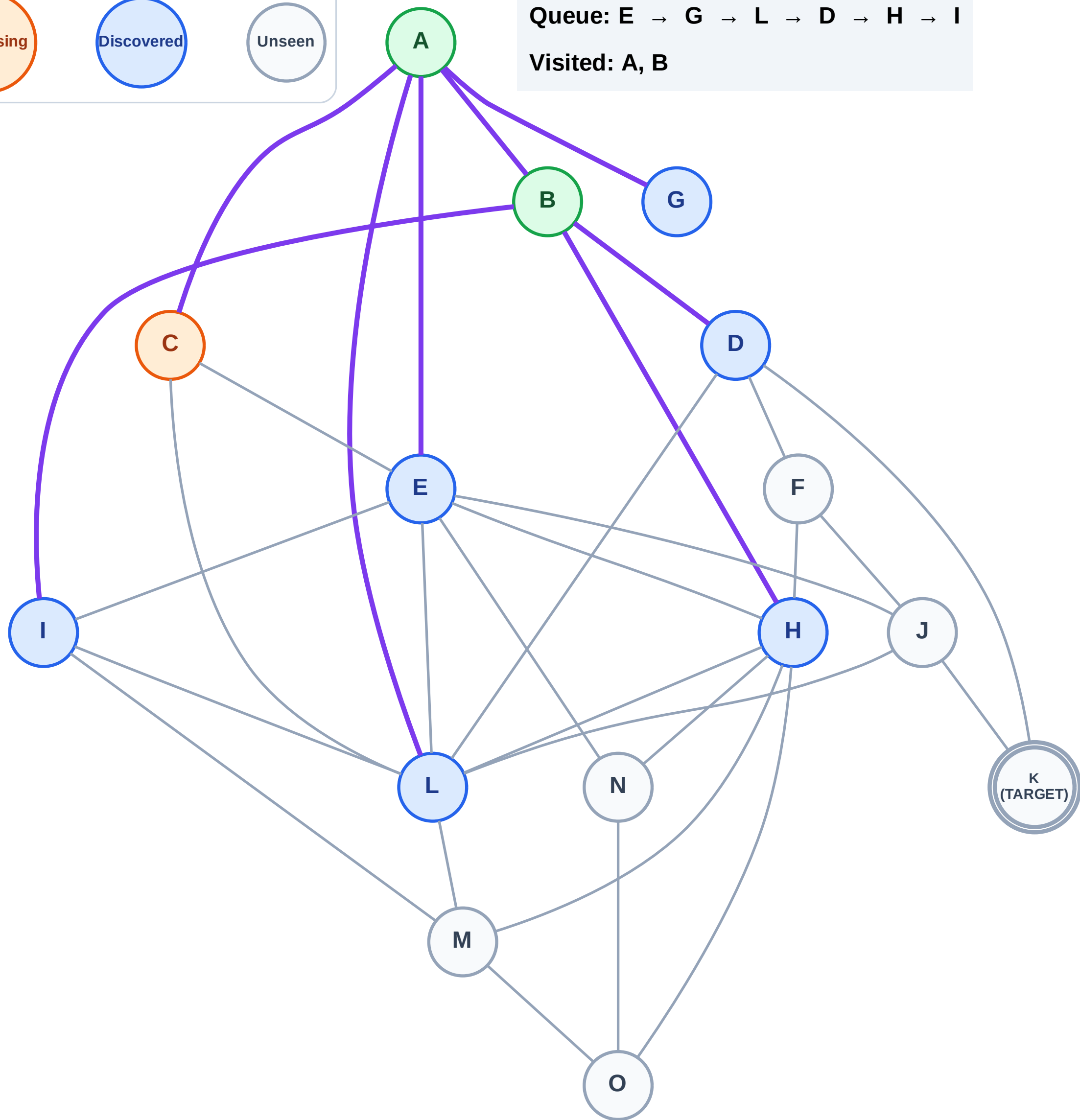
Processing

Discovered

Unseen

Queue: E → G → L → D → H → I

Visited: A, B



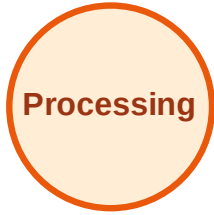
BFS Traversal

Step 7: No new neighbors from C

Legend



Complete



Processing



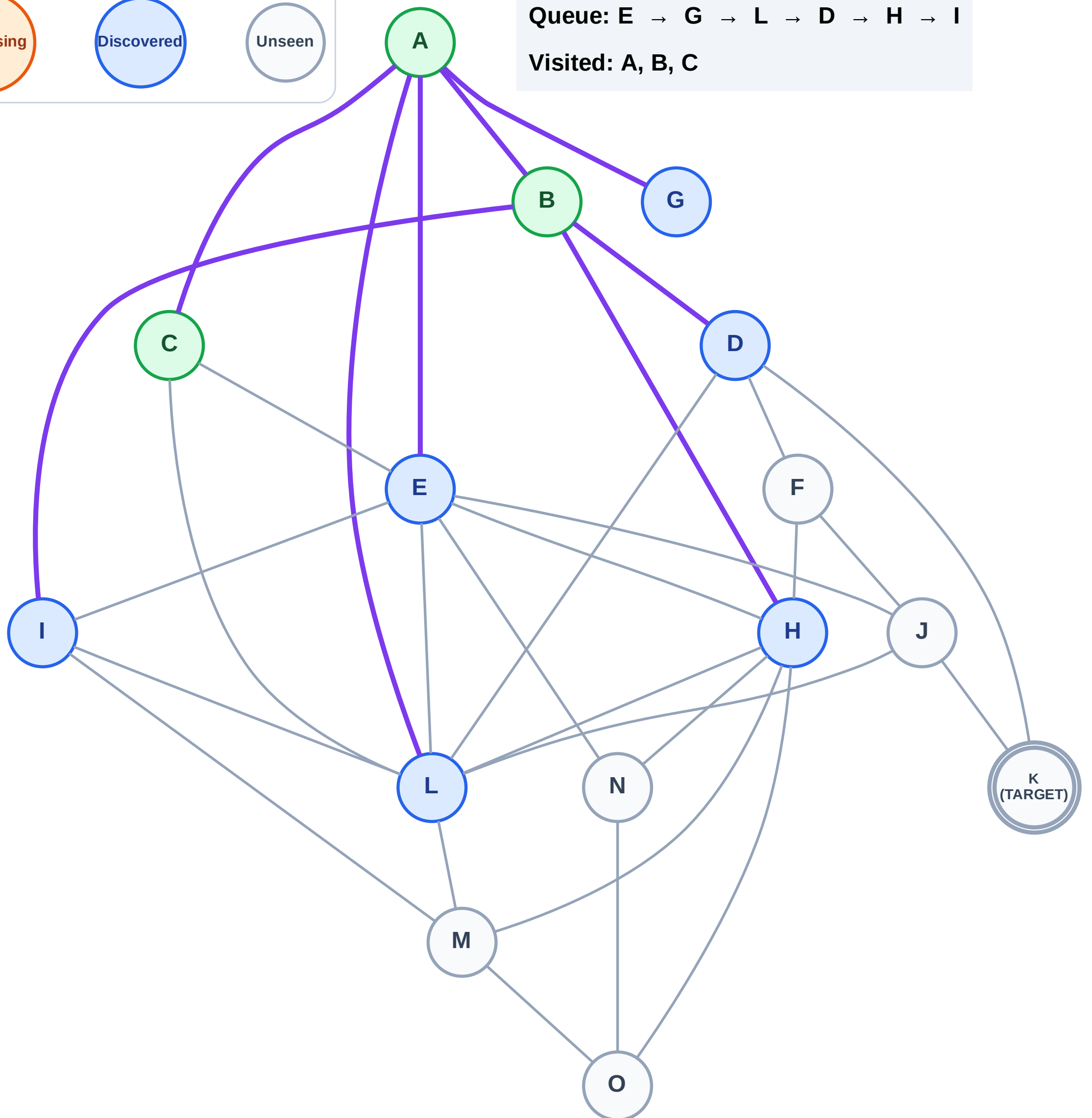
Discovered



Unseen

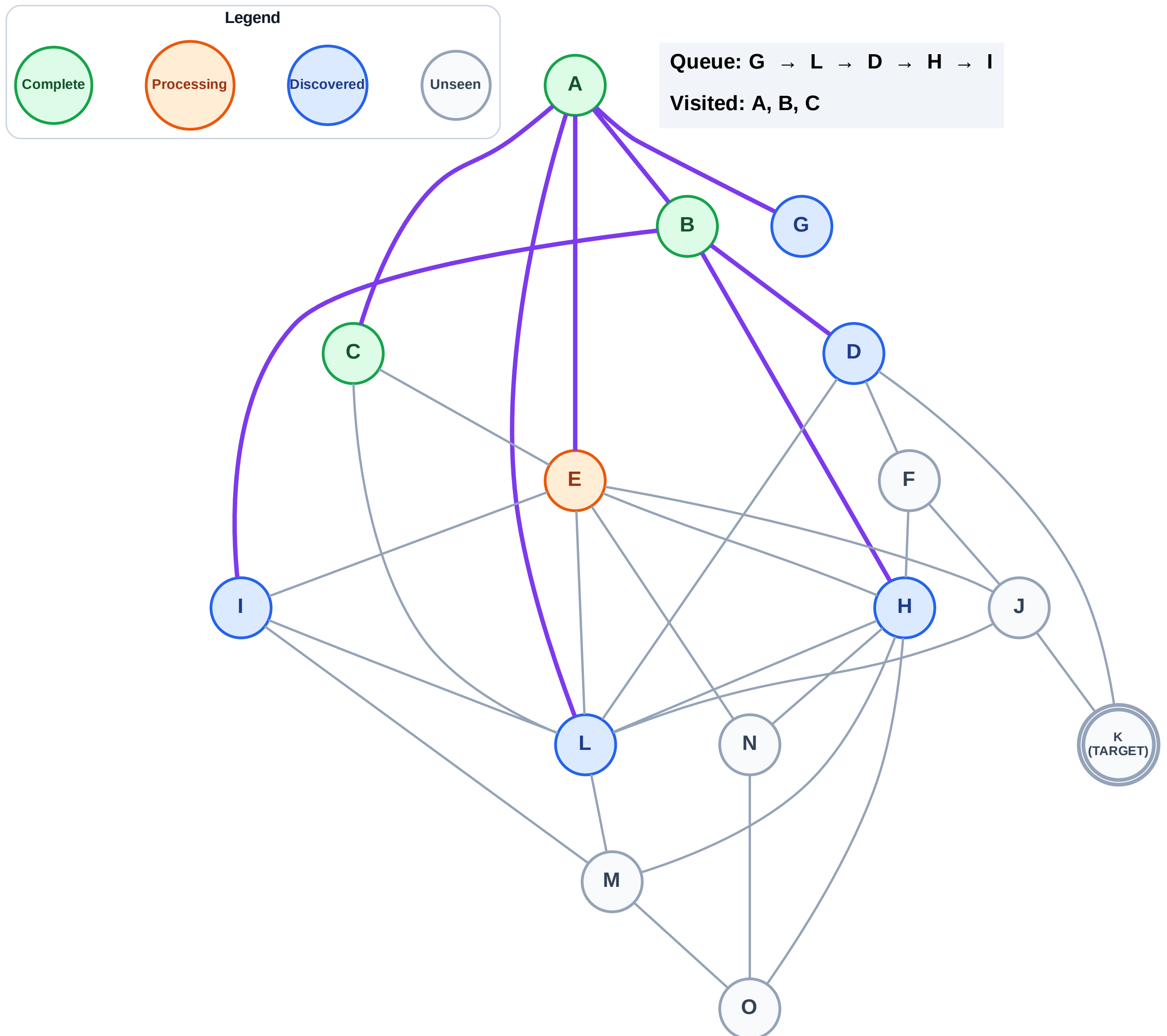
Queue: E → G → L → D → H → I

Visited: A, B, C



Step 8: Process node E

Step 8: Process node E



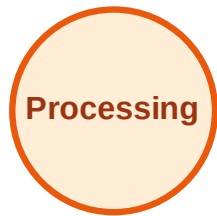
BFS Traversal

Step 9: Discover J, N from E

Legend



Complete



Processing



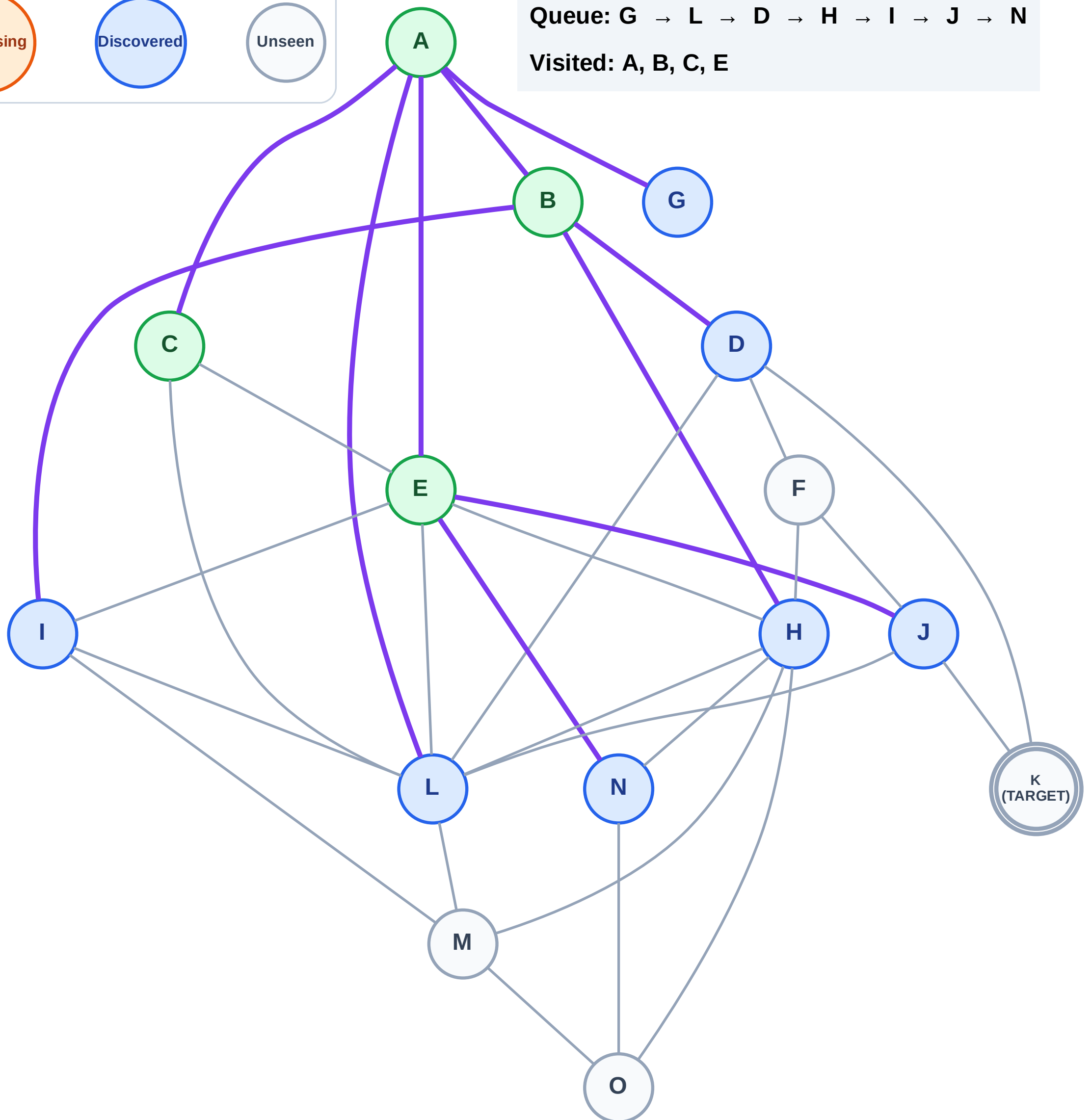
Discovered



Unseen

Queue: G → L → D → H → I → J → N

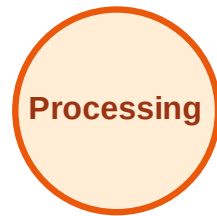
Visited: A, B, C, E



BFS Traversal

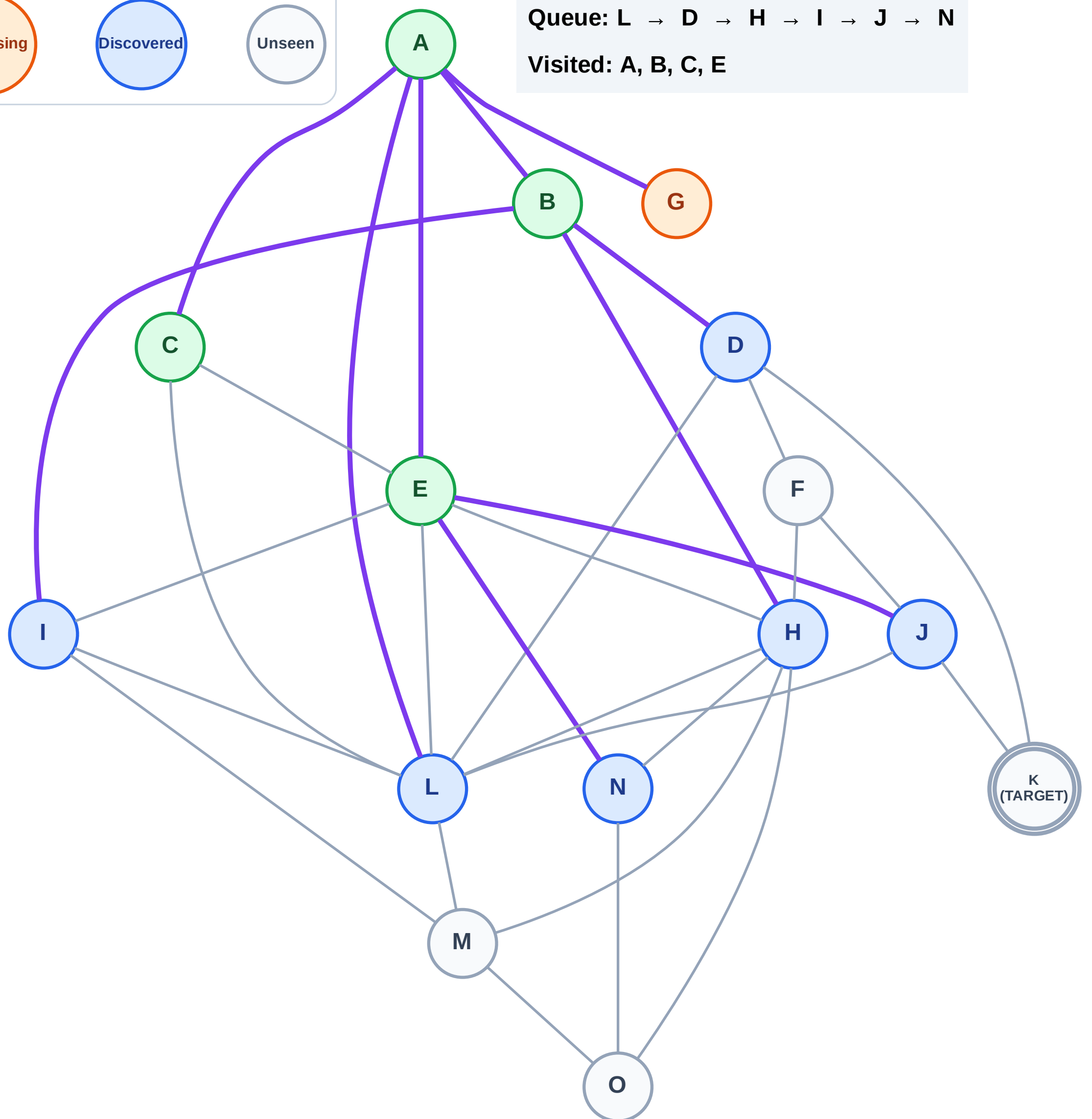
Step 10: Process node G

Legend



Queue: L → D → H → I → J → N

Visited: A, B, C, E



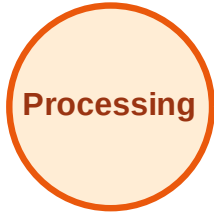
BFS Traversal

Step 11: No new neighbors from G

Legend



Complete



Processing



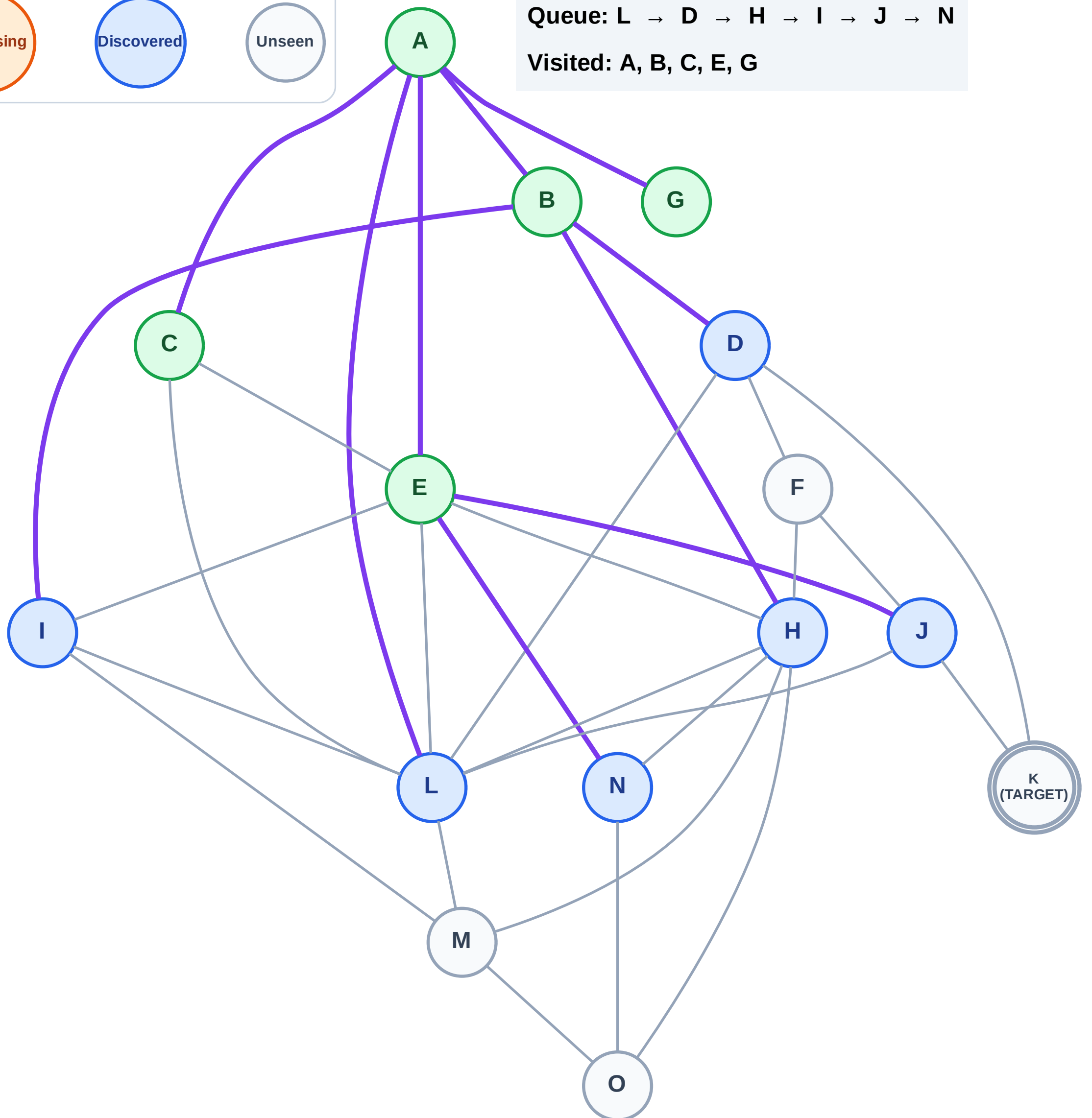
Discovered



Unseen

Queue: L → D → H → I → J → N

Visited: A, B, C, E, G



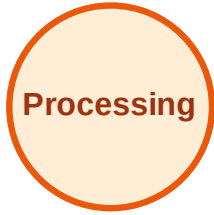
BFS Traversal

Step 12: Process node L

Legend



Complete



Processing



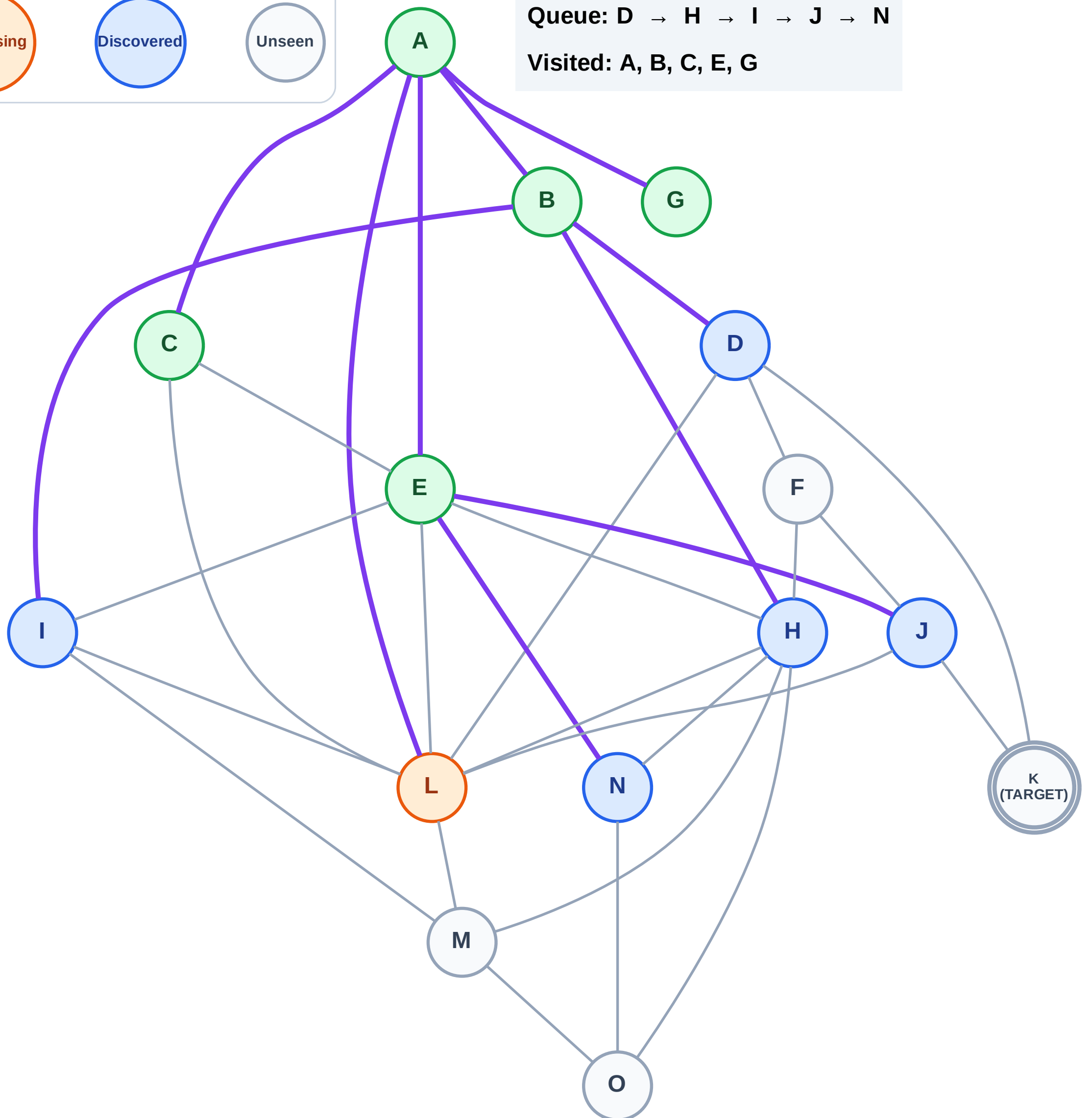
Discovered



Unseen

Queue: D → H → I → J → N

Visited: A, B, C, E, G



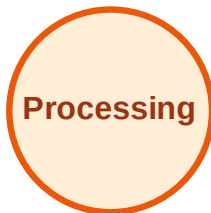
BFS Traversal

Step 13: Discover M from L

Legend



Complete



Processing



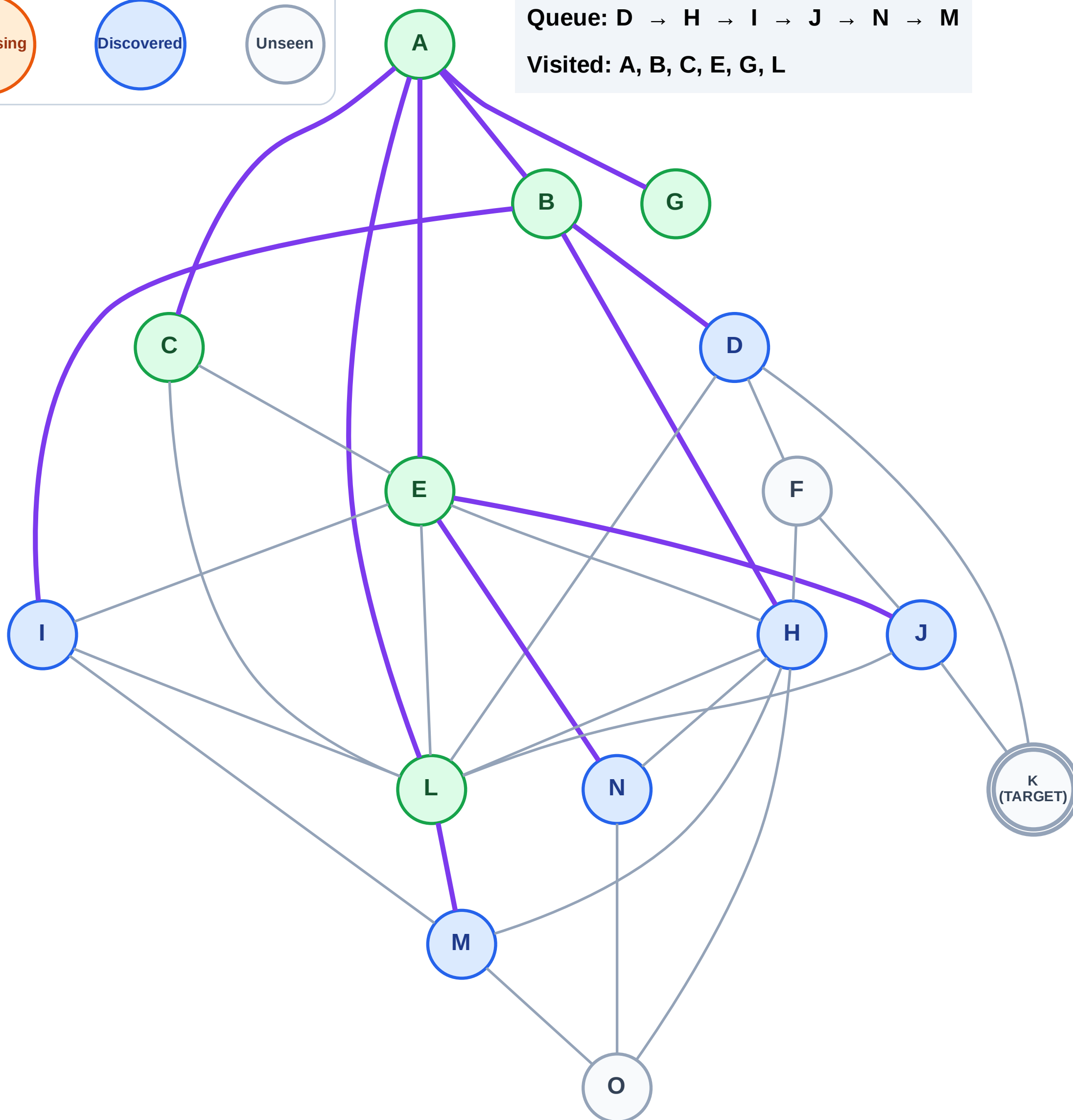
Discovered



Unseen

Queue: D → H → I → J → N → M

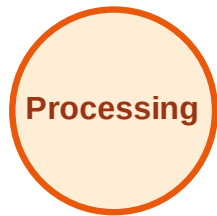
Visited: A, B, C, E, G, L



BFS Traversal

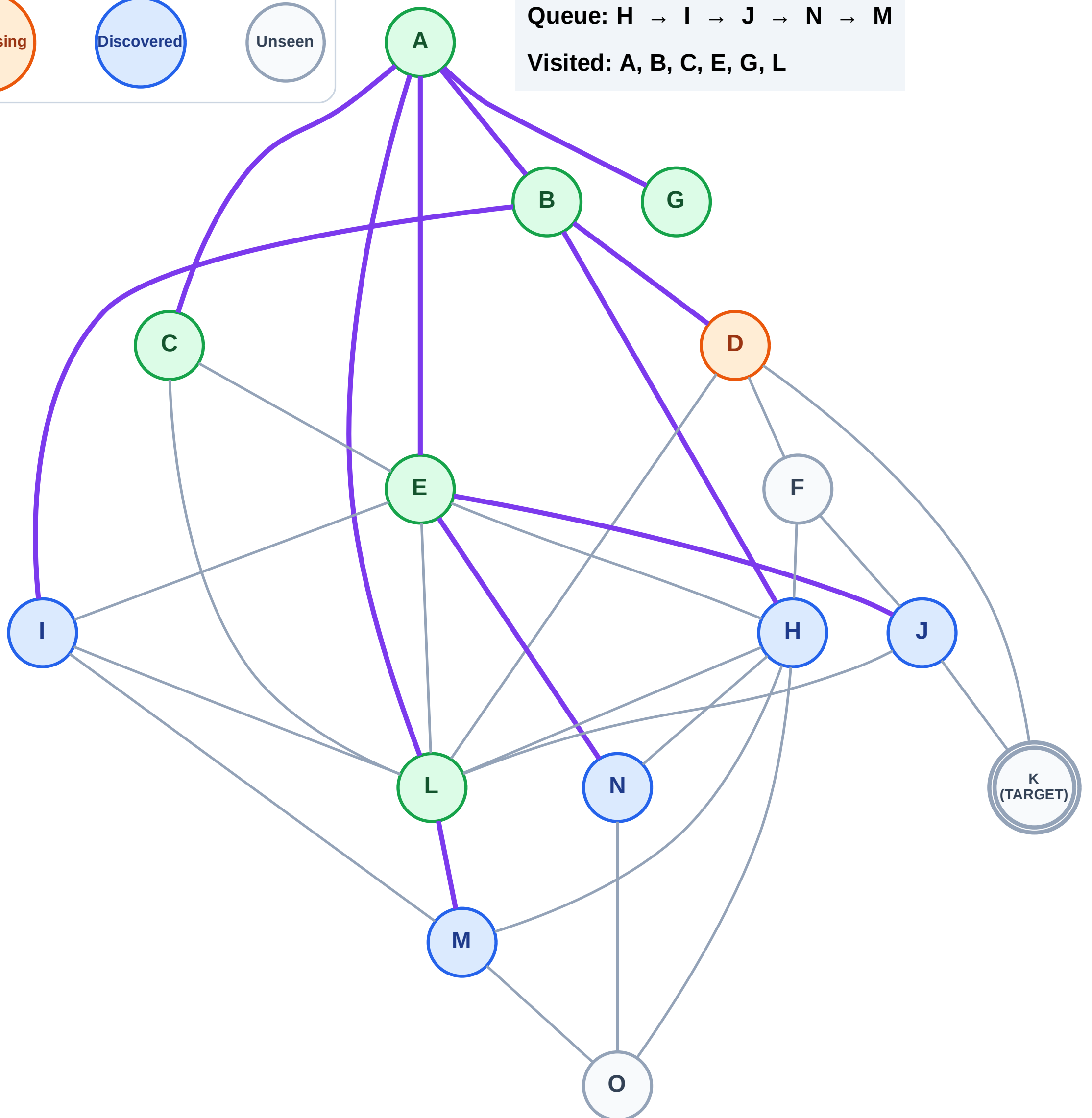
Step 14: Process node D

Legend



Queue: H → I → J → N → M

Visited: A, B, C, E, G, L



BFS Traversal

Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → D → K

Legend

Complete

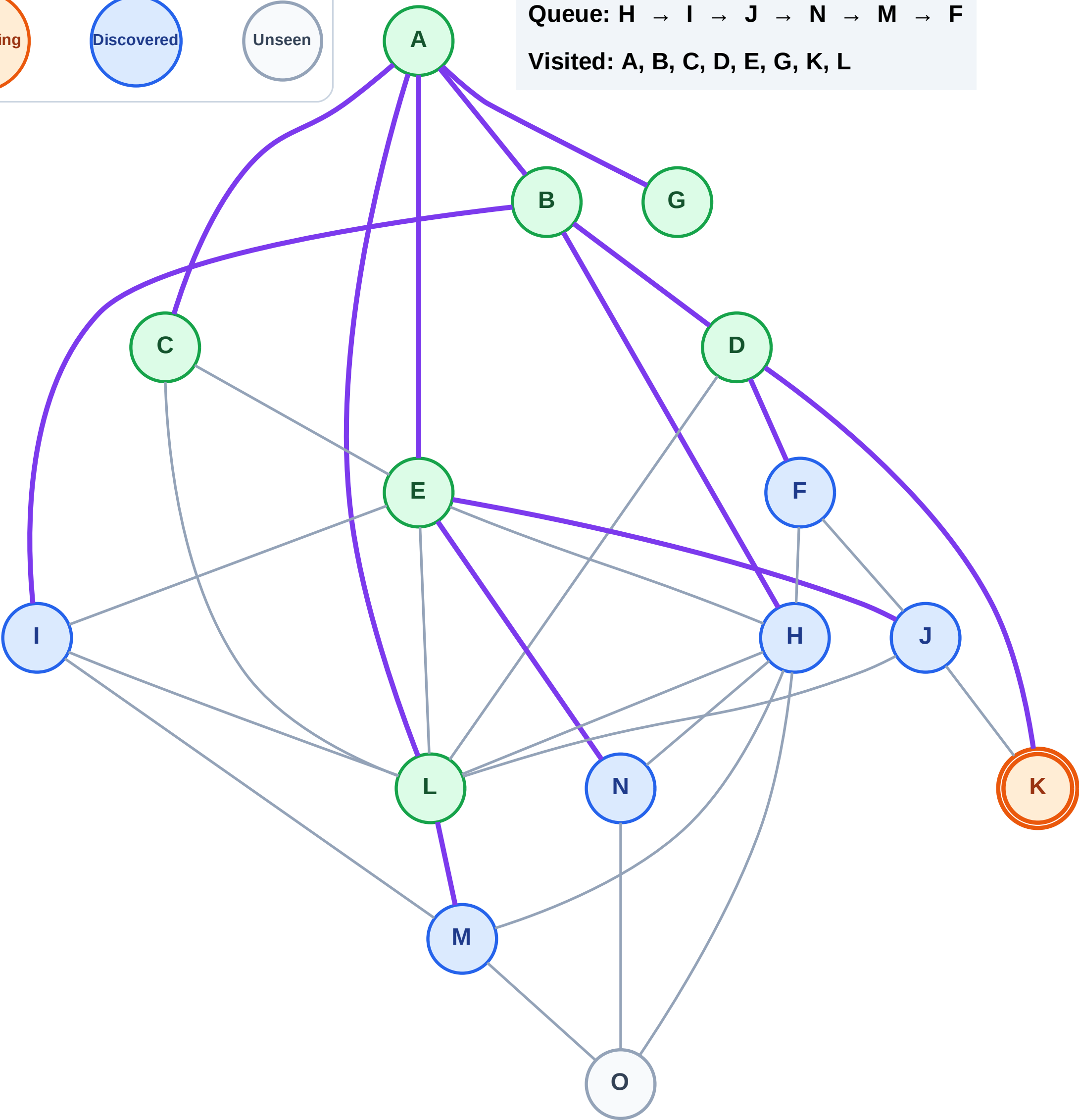
Processing

Discovered

Unseen

Queue: H → I → J → N → M → F

Visited: A, B, C, D, E, G, K, L



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Complete

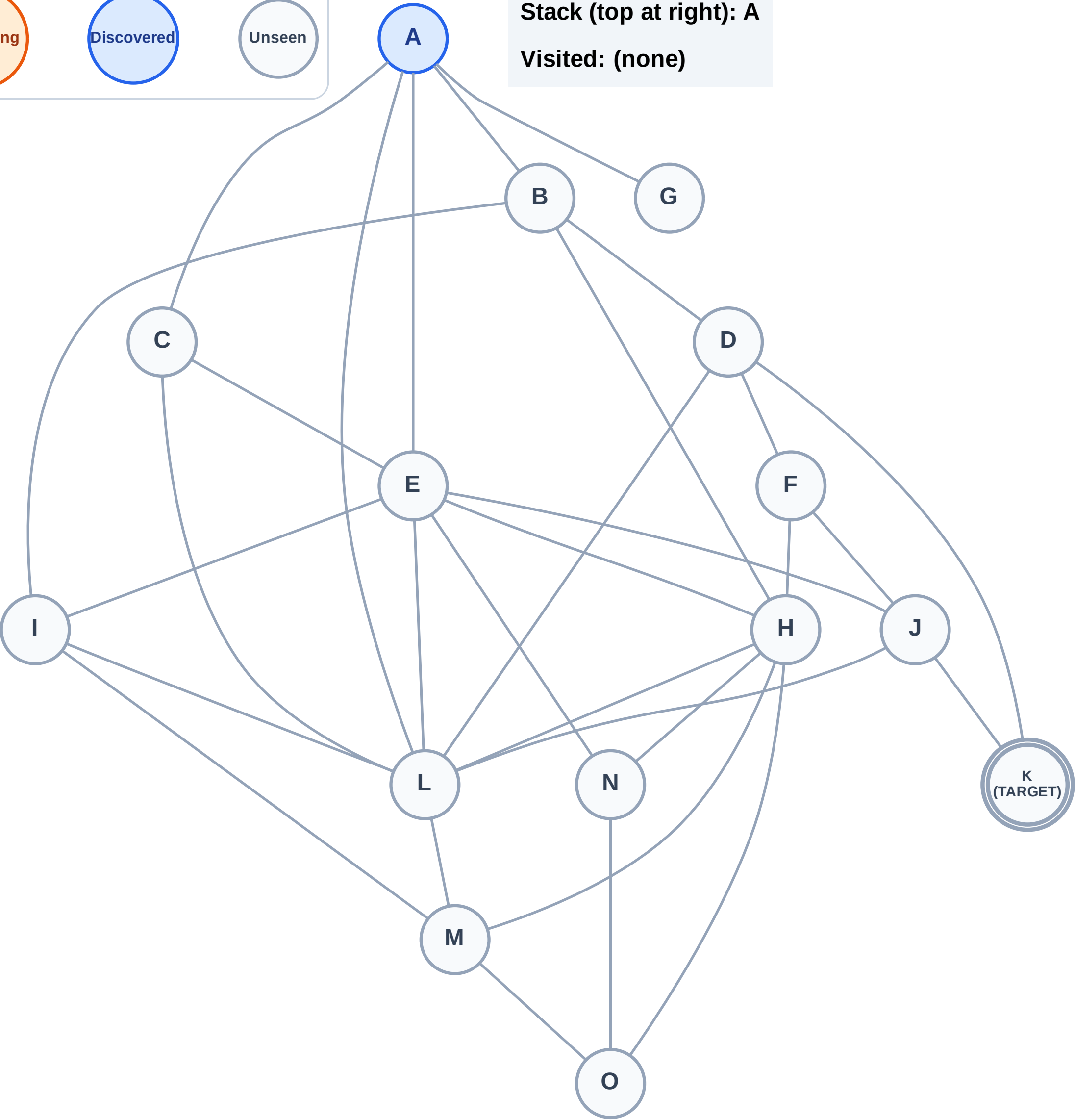
Processing

Discovered

Unseen

Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

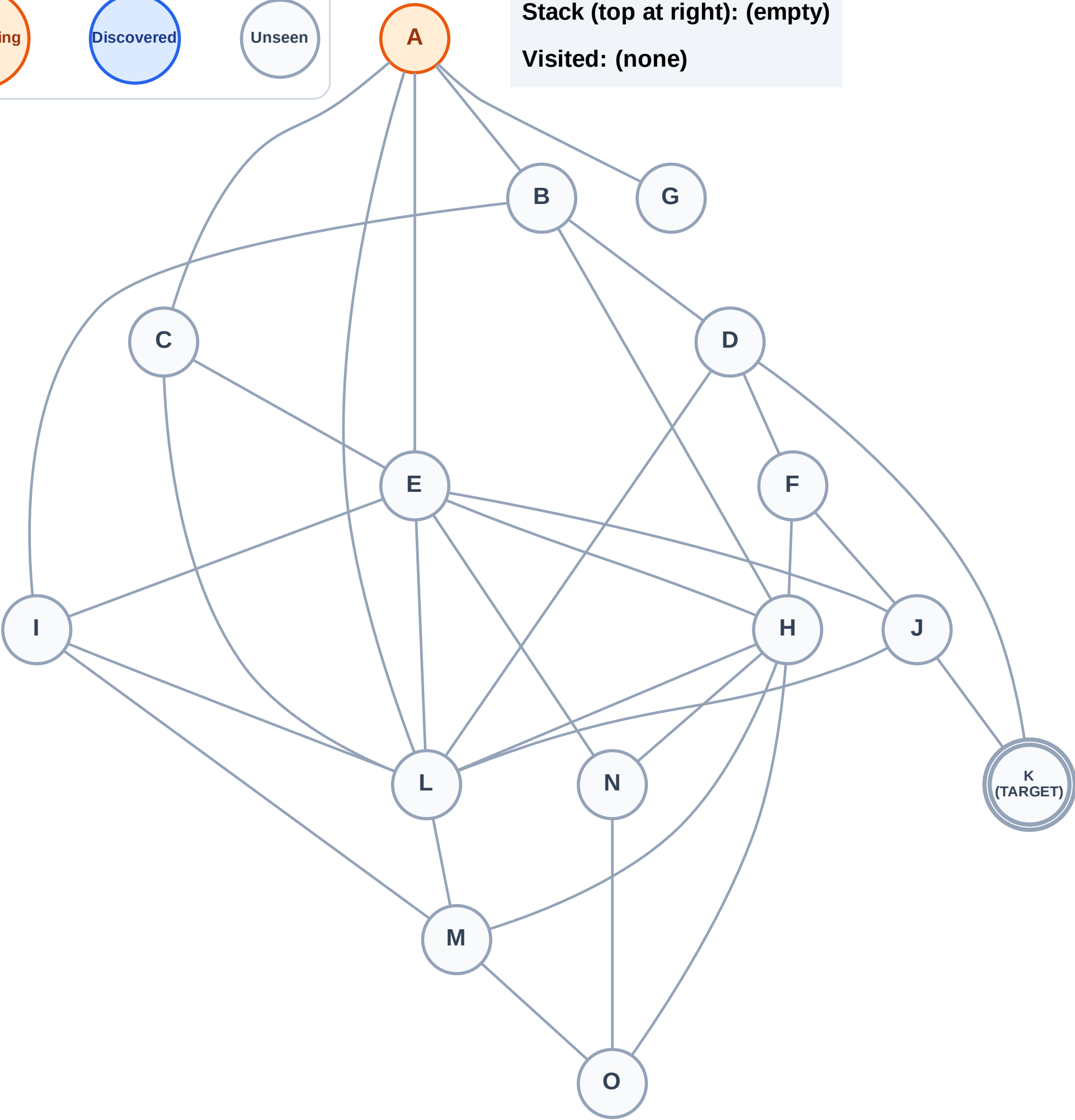
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



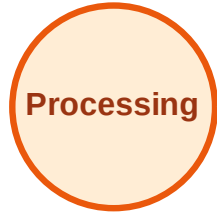
DFS Traversal

Step 3: Discover L, G, E, C, B from A

Legend



Complete



Processing



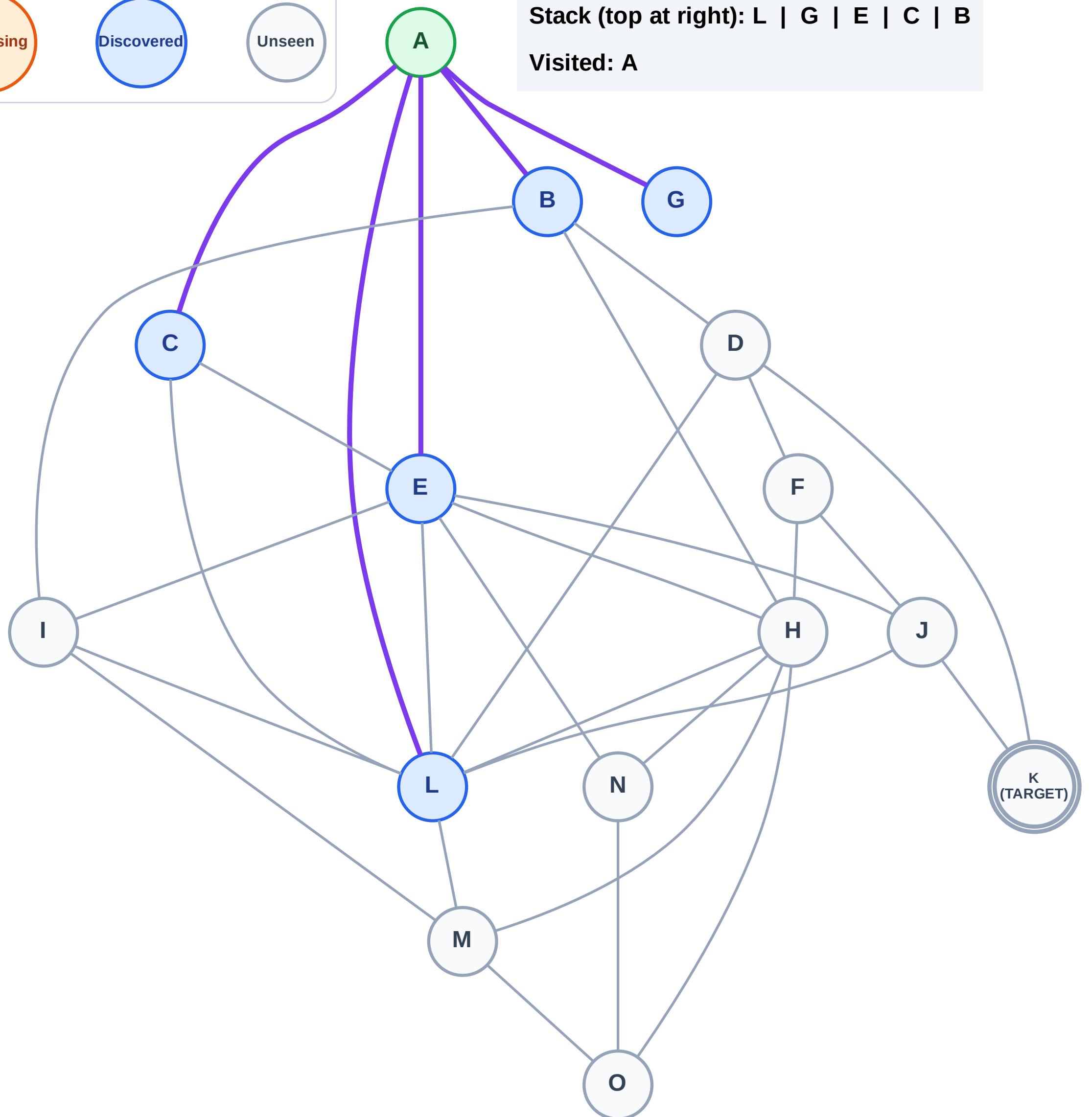
Discovered



Unseen

Stack (top at right): L | G | E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

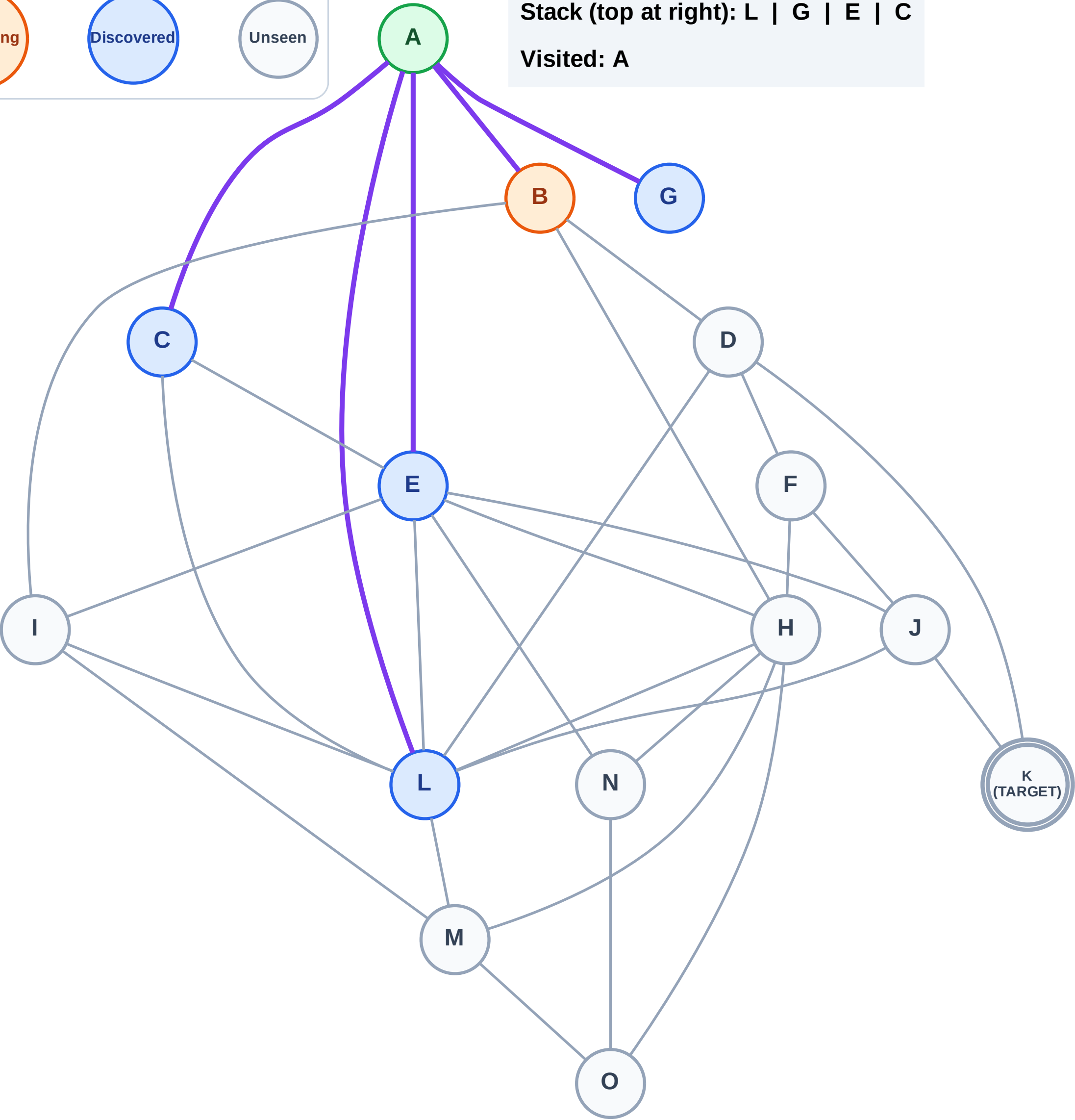
Processing

Discovered

Unseen

Stack (top at right): L | G | E | C

Visited: A



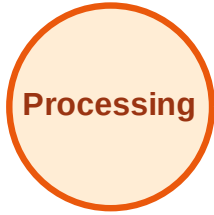
DFS Traversal

Step 5: Discover I, H, D from B

Legend



Complete



Processing



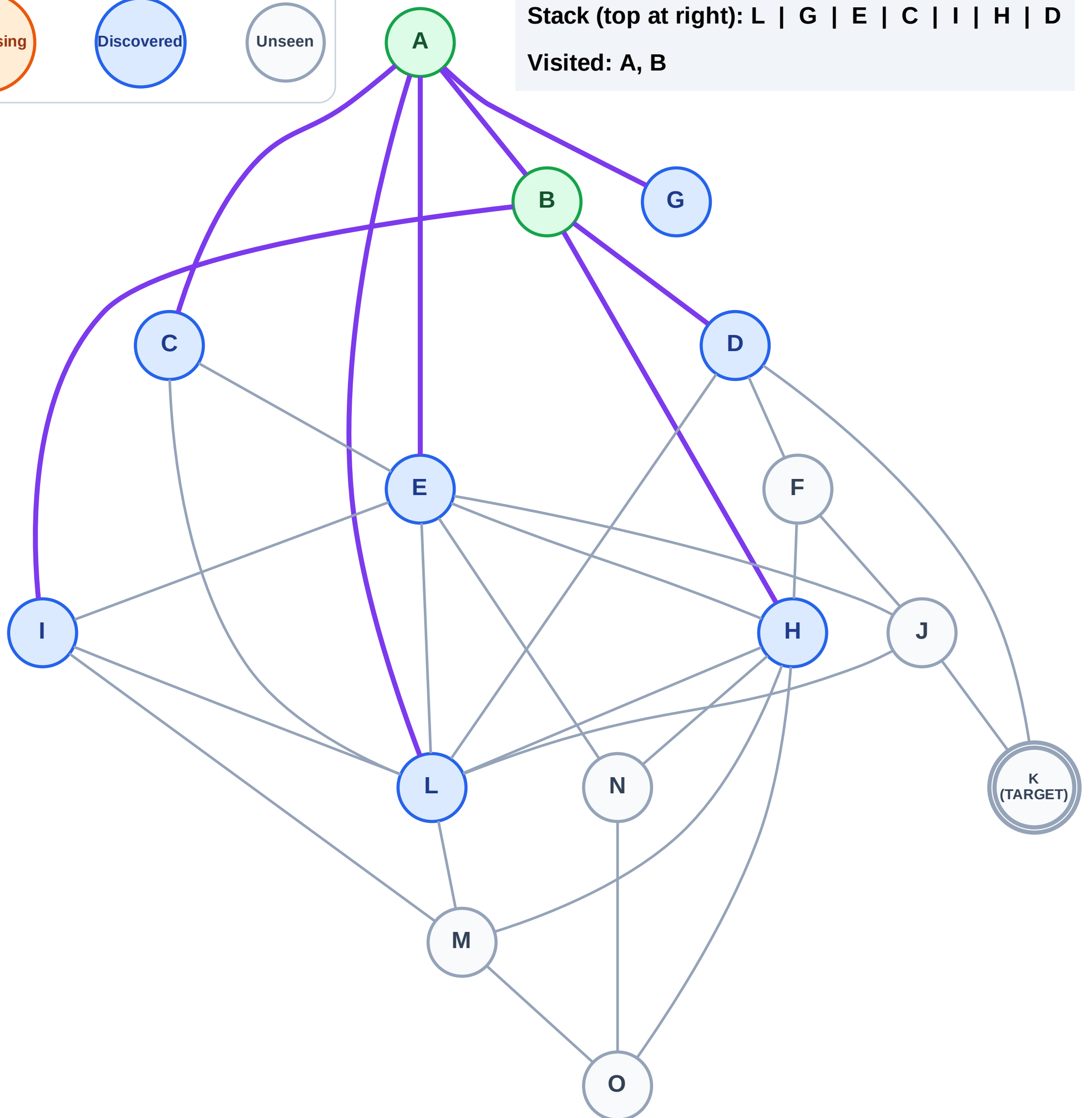
Discovered



Unseen

Stack (top at right): L | G | E | C | I | H | D

Visited: A, B



DFS Traversal

Step 6: Process node D

Legend

Complete

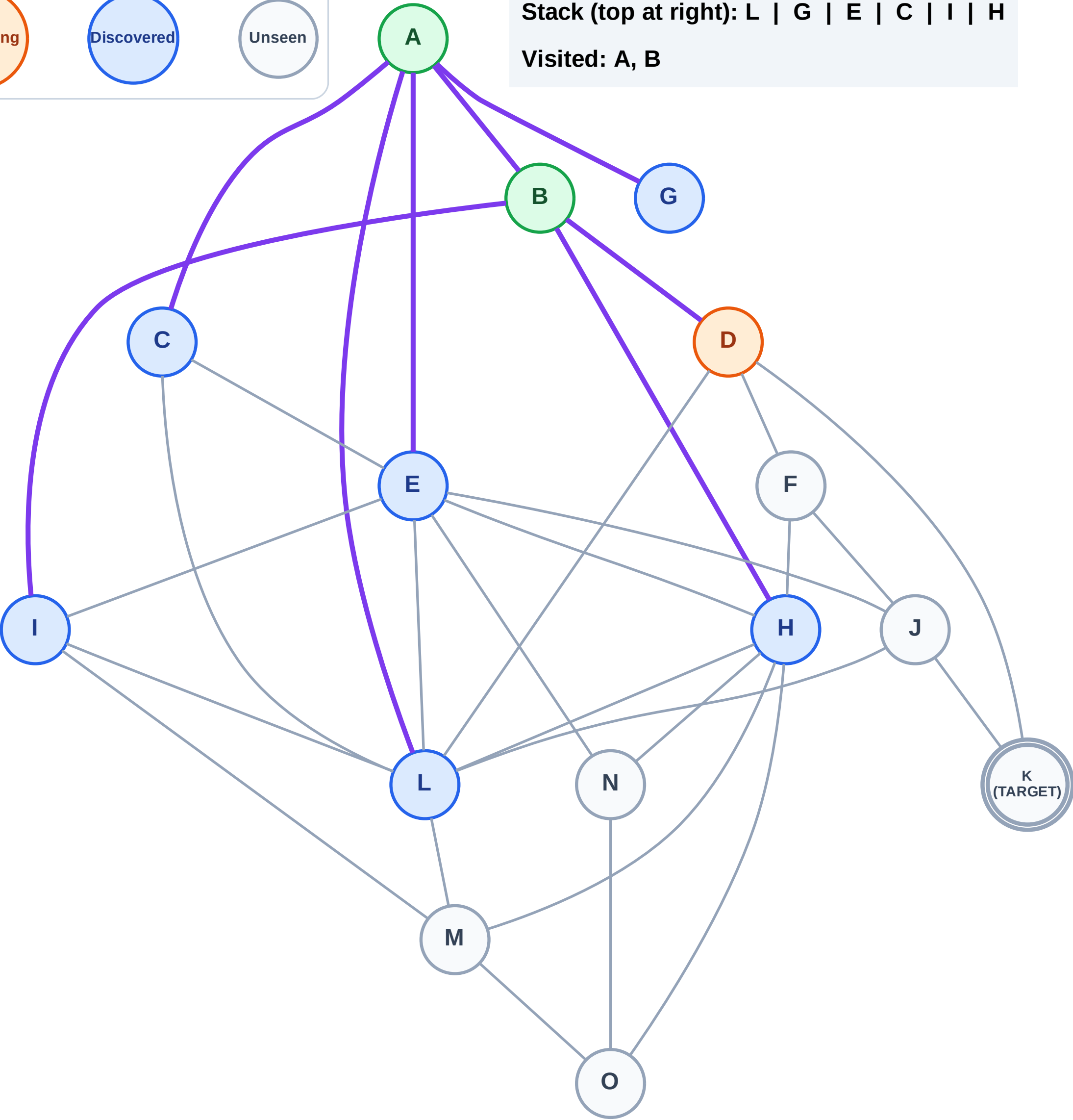
Processing

Discovered

Unseen

Stack (top at right): L | G | E | C | I | H

Visited: A, B



DFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → D → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): L | G | E | C | I | H
Visited: A, B, D, K

